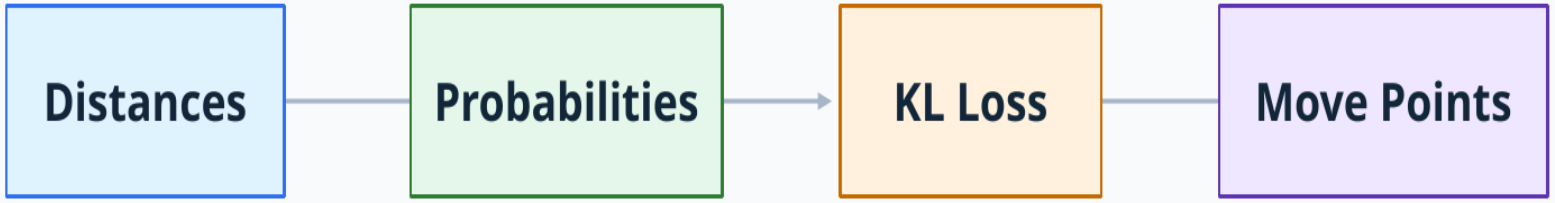
t-SNE Flow

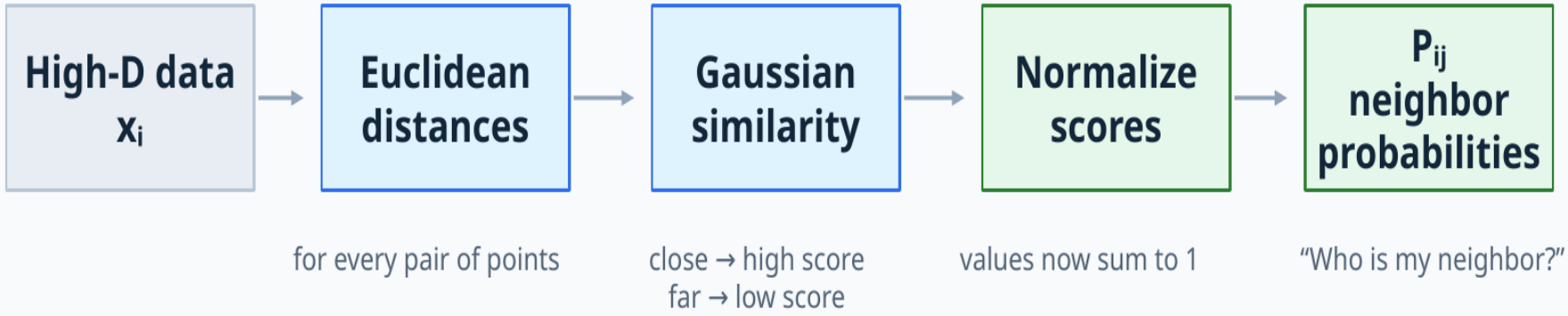
High-dimensional data → 2D/3D visualization



Flow only: the algorithm steps from input data to final map

1. High-dimensional probabilities

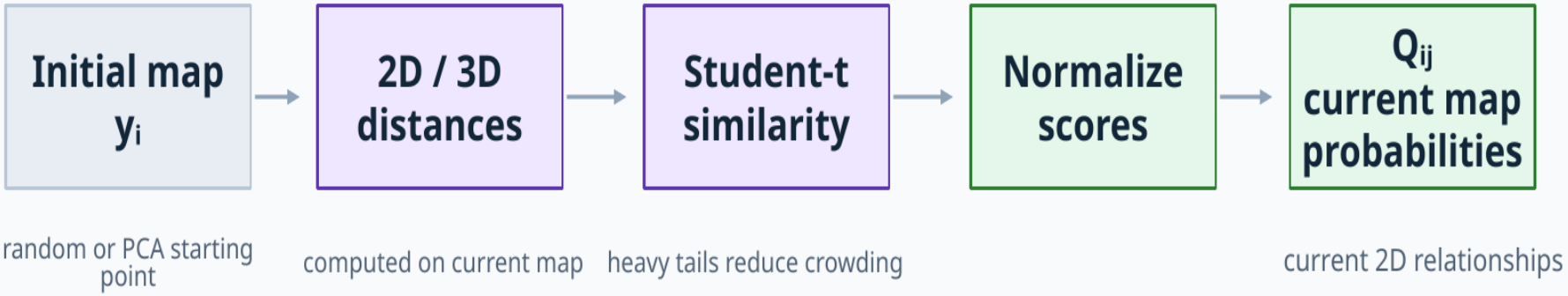
Goal: convert original distances into P_{ij}



High-D flow: $x_i \rightarrow ||x_i - x_j||^2 \rightarrow \exp(-d^2 / 2\sigma_i^2) \rightarrow$ normalize $\rightarrow P_{ij}$

2. Low-dimensional probabilities

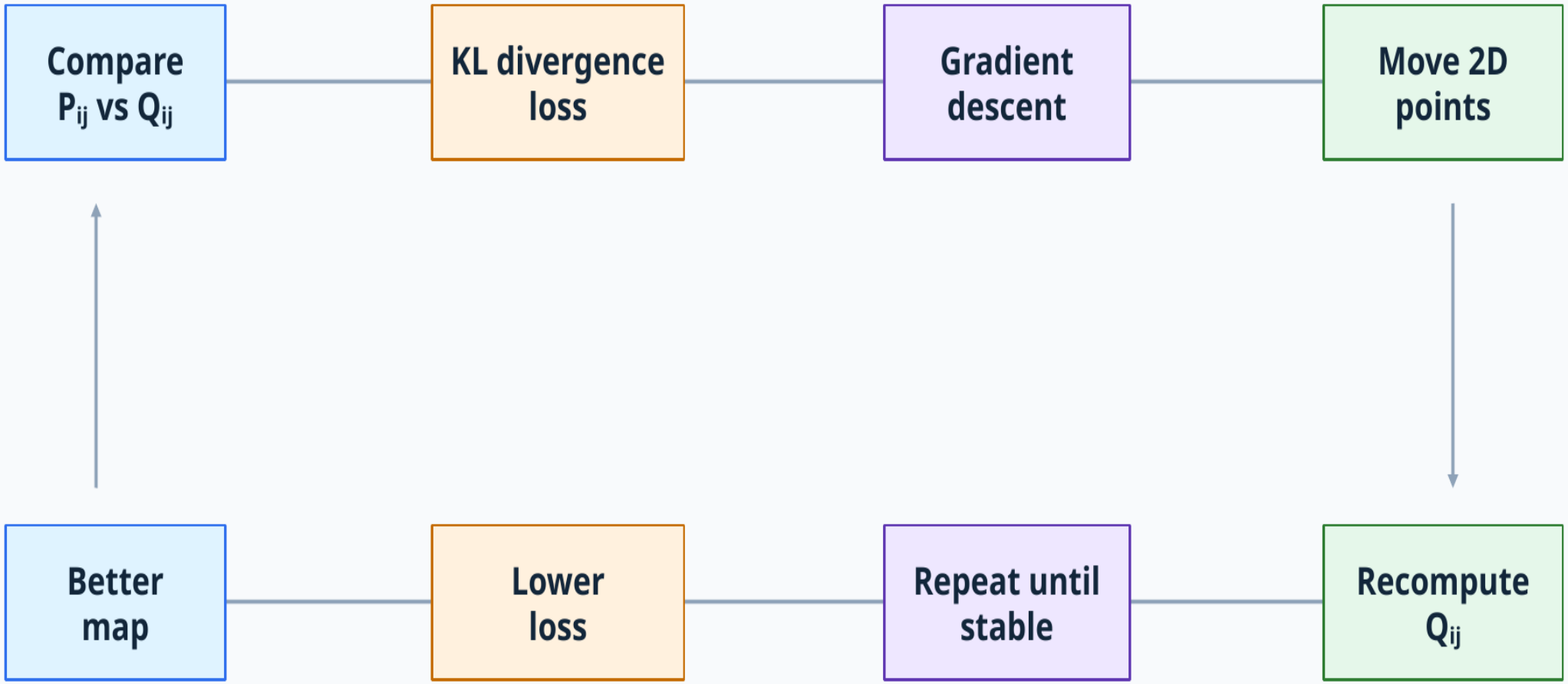
Goal: convert current map distances into Q_{ij}



Low-D flow: $y_i \rightarrow ||y_i - y_j||^2 \rightarrow 1 / (1 + d^2) \rightarrow$ normalize $\rightarrow Q_{ij}$

3. Minimize mismatch and move the points

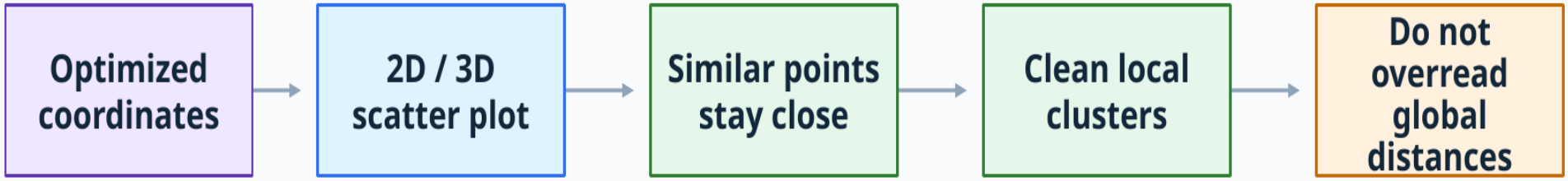
Goal: make Q_{ij} match P_{ij} as closely as possible



Optimization flow: P vs Q → KL(P||Q) → gradient descent → update y_i → repeat

4. Final t-SNE output

Goal: a 2D / 3D plot where local neighbor relationships are visible



t-SNE preserves local neighbor probabilities, not exact original distances.

Final interpretation: nearby dots are usually similar; distances between far clusters are not exact measurements.